JOSHUA HEAGLE

Senior Full-Stack Engineer

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Summary

Senior full-stack engineer with 12+ years shipping production web applications in PHP/Laravel, TypeScript/Node.js and Vue.js, with a consistent pattern of turning one-off solutions into shared, maintainable architecture. AI-native in daily practice: built a multi-agent development system (implementer and reviewer agents in sandboxed containers, an automated test gate, routing between Claude and local models) and ships with Claude Code across nearly 20 repositories, while owning the verification of every change. Maintains a TypeScript open-source ecosystem of 9 npm packages, including a SQL-like query engine for JSON data. Retained through a major business restructuring as one of four core engineers building BIC's in-house supply-chain platform.

Skills

Languages & Frameworks: TypeScript, JavaScript, PHP 8, Laravel, Node.js, Vue.js, React, HTML5, CSS3/Sass, Python (prototyping); WordPress, Drupal

Architecture & Patterns: REST APIs, webhooks and third-party integrations, event-driven design (Jobs/Events/Listeners on Redis queues), Strategy and Repository patterns, DTOs, shared package design

Data & Infrastructure: MySQL, MSSQL/T-SQL, Redis, Elasticsearch, Algolia, Docker, AWS

Testing & Delivery: PHPUnit, Jest, Laravel Dusk, Playwright, TDD, CI/CD (GitHub Actions), feature flags, release automation

AI Engineering: Claude Code, multi-agent orchestration, AI code-review workflows, local LLMs (Ollama), prompt-injection and least-privilege hardening

Leadership: release ownership, TDD adoption, mentoring developers, shared engineering standards across many repositories

Languages: English (native), Spanish (near-fluent), Mandarin (conversational)

Experience

Full-Stack Engineer — *BIC (formerly Inkbox), Toronto, ON* April 2021 – Present

- **Selected as one of four engineers (plus a QA) retained** to build in-house engineering capability on a newly formed Global Supply Chain team, after the consumer Inkbox business closed and BIC chose to invest in an internal team rather than continue relying on third-party contractors.
- **Designed and built an Inventory & Distribution system from scratch** — data model, schema, admin tooling, and integration with the platform's DTO layer — covering a catalog of tens of thousands of SKUs; hardened it through a real production bug-fix cycle.
- **Refactored a 1,489-line monolithic sheet-generation service** into a Strategy-pattern architecture (a shared interface, six generator types, composed traits), cutting the core class to 128 lines; later extended the same pattern to support bundled products and added caching for performance.
- **Introduced the platform's Data Transfer Object pattern** and maintained it for years; drove its extraction into a standalone shared package now used across multiple codebases.
- **Built and maintained the discount system and product feed pipeline** (spanning both the core backend and the customer-facing storefront) over multiple years each; shipped storefront features behind feature flags, occasionally exposing them live to gauge consumer response.
- **Built asynchronous, event-driven workflows** with Laravel Jobs, Events, Listeners and Observers on Redis and database queues, alongside GitHub Actions CI.
- **Prototyped and benchmarked a sort/packing algorithm** for a roll-to-roll print-sheet generator, choosing PHP's built-in usort over a custom implementation after measuring both under real throughput constraints; separately experimented with a Python demand and sales forecasting prototype for production planning.

- **Broad platform ownership:** search infrastructure (Elasticsearch/Algolia), analytics event tracking, third-party webhook integrations (Shopify, BlueTree), Vue.js frontend work, Laravel Dusk end-to-end tests, and internal admin tooling.

Software Developer II (promoted from Technical Analyst) — *Benevity, Toronto, ON* January 2017 – April 2021

- **Architected and built new API services** as part of a small team; built the Grants Forms service from the ground up, including a customizable adapter for routing submissions to arbitrary receiving APIs.
- **Owned the team's weekly release process end-to-end:** triaged tagged Jira tickets, built and synced Docker images with AWS, ran the accompanying MSSQL scripts, coordinated with QA to validate the release, and stayed ready to roll back to a prior image if anything broke.
- **Introduced a test suite and TDD tooling** on a team recognized internally as the company's best-performing — regularly consulted by larger, less-efficient teams for advice on improving their own process.
- **Helped cut the test-suite run time by two-thirds** (from up to 15 minutes to 5) by finding the slow-downs and reworking legacy tests to current code standards.

Earlier Experience

Intermediate Web Developer, BID13 (2016–2017) — Drupal migrations and custom modules, event-driven email and alert triggers, mentored junior developers.

Developer / Systems Analyst, Techcom (2014–2016) — Led client-facing development, coordinated a project team on secure financial applications.

Front-End / PHP Developer, 3conx & Jackpine (2013–2014) — Client websites built from designers' mock-ups; responsive, cross-browser front ends.

Selected Open Source & AI Projects

- **agent-env — multi-agent development system:** implementer and reviewer agents run in sandboxed Docker containers under their own GitHub App identity, gated by build, test and lint checks; tasks are routed between Claude and local Ollama models by a classifier evaluated on a test set; run records track cost and outcome; hardened against prompt injection with least-privilege, short-lived tokens.
- **json-fs-query (npm) —** a SQL-like query language and engine for JSON file databases: hand-written parser, schema statements, indexes and typed ordering, released through 2.x with bugs caught by its own test suite before release.
- **js-build-tools (npm) —** the shared build, test and documentation pipeline (Gulp, Babel, TypeScript, Jest, Playwright, TypeDoc/JSDoc) behind nearly 20 of my repositories; removing an unneeded polyfill dependency cut browser bundles by about 55% across 14 projects.
- **json-dom, pseudo-dom (npm) and matrix-dom —** strict-TypeScript libraries that render typed, JSON-described component trees into the DOM (server-driven UI), provide a lightweight DOM for testing, and lay out grids; json-dom-components adds 32 reusable UI components. Used to build a multiplayer Battleship game synced in real time over Socket.IO.

Education

Algonquin College — Advanced Diploma, Computer Engineering Technology – Computing Science (2010–2013)